

Woolworths Online Order and Delivery System – Enterprise Systems Analysis, Design, and Modelling

Shawn Lee (23204035)

Table Of Content

Introduction	3
Organisational and System Analysis	3
Organisation Overview	3
Departmental Structure and Responsibilities	4
Problem Statement	5
Stakeholders and User Roles	8
Business Goals	9
Information from Stakeholders	9
User Requirements	9
System scope, boundaries, and assumptions	9
Requirements	9
Requirement UML Modelling	9
Initial Structural Design and Implementation	10
Conclusion	10
References	10
Appendices	11

Introduction

- ☐ Brief intro to the project
- ☐ Purpose of the report
- ☐ Overview of what each section covers

Organisational and System Analysis

- ☒ 2.1 Organisation Overview
- ☒ 2.2 Departmental Structure and Responsibilities
- ☒ 2.3 Problem Statement
- ☐ 2.4 Stakeholders and User Roles
- ☐ 2.5 System Scope, Boundaries, and Assumptions

Organisation Overview

Woolworths New Zealand (rebranded from Countdown) is a subsidiary of Woolworths Group Limited, an Australian publicly listed company. Woolworths New Zealand is a major supermarket chain whose headquarters are located in Mangere, Manukau, Auckland. Woolworth Group was founded by Percy Christmas on September 22, 1924, in Sydney, Australia. It is one of the 3 major supermarkets (Pak'n Save, Woolworths, New World) in New Zealand. Currently, there are 185 supermarkets^[1] in New Zealand and 20,000 employees^[1] working in New Zealand.

Woolworth's core values are providing convenience, value, range, and quality, and a motto of "The Fresh Food People"^[2]. Thus, Woolworths needs dependable information systems to manage its operations across shops, distribution centers, and digital platforms because it depends greatly on customer satisfaction. One of these operations that would be focused on in this report is the Online and Delivery System, since it offers greater convenience and the ability to "save time by ordering groceries online.

These advantages in these systems align with the organisation's values, in which the main focus is to provide the best experience for customers.

Departmental Structure and Responsibilities

In the ordering and delivery systems in Woolworths New Zealand, there are 4 critical departments in which forms the operation of the system. These systems are Store Operations, Supply Chain & Logistics, Customer Service, and IT.

- Store Operations

The store operations department for Woolworths New Zealand in 185 supermarkets^[1] focuses on restocking shelves and maintaining the food freshness of merchandise. In the context of an online ordering and delivery system, they are responsible for managing orders online by picking and packing them for delivery.

- Supply Chain & Logistics

The Supply Chain & Logistics department for Woolworths New Zealand focuses on managing stock movement from the supplier and distribution centres to the store. In the context of an online ordering and delivery system, they are responsible for assigning deliveries to complete orders and managing their delivery schedule, as 22%^[3] of orders online are fulfilled within 2 hours, and 57%^[3] of orders online are fulfilled within the same day of ordering, as well as ensuring there is enough stock for deliveries.

- Customer Service

The Customer Service department for Woolworths New Zealand focuses mainly on making sure that all customers are satisfied with the service and merchandise by handling customer inquiries, complaints, and issues. In the context of an online ordering and delivery system, this department is responsible for handling customer complaints and issues, assisting customers online, and processing refunds for incorrect or out-of-stock merchandise on the online ordering and delivery system.

- IT

The IT department for Woolworths New Zealand focuses on maintaining all digital platforms and technology systems, ensuring that user data is secure, developing improvements to the website, and resolving technical issues. In terms of the context of the online ordering and delivery system, the IT department is responsible for maintaining the system, integrating inventory, supply chain, and payment systems.

- People and Culture (HR)

The People and Culture Department for Woolworths New Zealand focuses on making sure that all employees are well-trained, supported in their well-being, and manage payroll and recruitment. In the 2025 annual report, mental health programs such as 'Sonder' and leadership skills are completed by over 55,000^[3] employees. This department also manages wage agreements and ensures there is an inclusive environment and culture.

- Finance

The finance department for Woolworths New Zealand is involved in managing the budget, managing the cash flow, planning, and analysing financial decisions, analysing financial risk, as well as producing financial reports. These responsibilities are important in overseeing the financial report of Woolworths New Zealand. Food's total sales reached \$8,286 million^[3] with normalised sales.

- Marketing

The marketing department for Woolworths New Zealand's main goal is to build brand awareness, build brand loyalty through rewards, and research the trends of the market, creating promotions and advertisements. This includes the marketing department managing the grocery shopping website, which contributed to having 2.1 million^[3] active everyday reward members in Q4 of 2025.

Problem Statement

- ☐ Creating a problem with the existing system could improve the system

- ☐ If there are no improvements that could be made, then you just recreate the existing system
- ☐ Need to improve on why the system is needed
- ☐ Need to describe both the customer and the business

As online shopping continues to grow in popularity due to its convenience, Woolworths New Zealand has identified a need to implement an online system of its own. Currently, Woolworths New Zealand does not have a dedicated online order and delivery system, relying on manual in-store shopping as the primary means of purchasing groceries, and has no system in which to manage the delivery of products when customers order online. This limits customer convenience and restricts Woolworths New Zealand's ability to meet growing demand for online shopping.

Furthermore, Woolworths New Zealand does not have an online or delivery system. From the customer's perspective on the manual purchasing of groceries, there is no efficient way in which customers are able to compare and decide different product choices based on their nutritional value, price, etc. Customers would also not be able to access Woolworths online, as there is no system that provides the ability for customers to purchase goods. The customer would not have the convenience or lack the physical ability to enter or check which store has a special promotion, opening, or closing date, or any news about the organisation.

From the perspective of the store operation employees, they currently are not able to visualize the total number of products offered in the store, thus needing to manually record and manage product stocks, which could increase the chances of human error. Employees would not be able to have statistical data on which product has the most demand, which leads to poor positioning of where products should be stocked.

The lack of an online system in which provide analytical data would make them rely on previous data about the stock levels and not current, real-time demand data, thus providing inaccurate data to the procurement specialist and the inventory controller. This raises operating expenses and the possibility of stock imbalances by causing poor forecasts and ineffective supply allocation across various shop locations.

From the standpoint of customer service, the absence of an online platform implies that there is no system to process online refunds, no way to alert consumers of delivery updates, and no organised method to manage digital customer complaints. This makes it more difficult for Woolworths New Zealand to quickly address problems found by customers, which over time may erode consumer loyalty and confidence.

The absence of an IT system means Woolworths New Zealand's ability to reach and engage customers online is severely restricted, as there is no centralised infrastructure to support key business functions such as order management, customer data storage, delivery tracking, or complaint handling.

Woolworths would have a significant disadvantage competing with other organisation in the market who leverage customer data to deliver targeted promotions and personalised shopping experiences on their online system. The organization's long-term development potential is limited since it cannot scale its operations to satisfy the increasing demand for online grocery shopping without this digital structure.

Woolworths New Zealand needs to develop an Online Order and Delivery System that allows customers to place an online order, in which customers are allowed to browse and search for products online. The customer should also be able to add products to their shopping cart and be allowed to check out. Customers would also be able to save a shopping list to reuse previous lists of orders to make it faster to shop for weekly used products. Customers will be able to create an account so that their details and saved shopping list are stored on the account for ease of access. The second part is the delivery system, which helps automate delivery order schedules so that delivery employees can be assigned to a schedule based on their location and availability to maximise the efficiency of delivery. The system should allow drivers to update deliveries in real time so that customers can view and be notified when their orders will arrive. Store operation staff would need to be able to receive information about the order and pick it ready for delivery. Shift managers need to be able to assign store staff by using the online system to delegate tasks as well as monitor the operations of tasks.

The system should improve customer convenience, streamline order fulfilment, and provide Woolworths New Zealand with a scalable platform for online grocery shopping.

Stakeholders and User Roles

Department	Stakeholder	Role
IT	System Administrator	Manages system access, user accounts, and permissions
IT	IT Department/Support	Maintains and supports the system infrastructure
Store Operations	Store Operations Staff	Picks and prepares orders in the store
Store Operations	Store Manager	Oversees overall store operations and monitors system performance.
Store Operations	Shift Manager	Oversees store operations and staff delegation
Supply Chain and Logistics	Delivery Driver/Employee	Fulfils deliveries to customers
Supply Chain and Logistics	Procurement Specialist	Specialist Purchases stock for store locations
Supply Chain & Logistics	Inventory Controller	Manages stock distribution across store locations
Customer Service	Customer	Primary user of the online shopping platform
Customer Service	Customer Service Representative	Handles customer complaints and refunds

Business Goals

- Customer Service
 - Improve customer convenience by providing an accessible online grocery shopping platform.
 - Enhance customer satisfaction through effective delivery communication.
 - Improve customer trust through an accessible online complaint and refund platform.

- Store Operations
 - Decrease the amount of in-person communication that shift managers and retail employees use.
 - Reduce human error in store operations through digital task and order management.
 - Increase stock management precision with real-time stock visibility.

- Supply Chain & Logistics
 - Use automated delivery scheduling to increase delivery efficiency.
 - Incorporate real-time delivery status updates on deliveries.
 - Improve forecasting accuracy by providing procurement and inventory staff with real-time data demands.

- IT
 - Create a centralised digital infrastructure to assist with important company operations.(order management, customer data storage, and delivery tracking).
 - Allow the system's online ordering, retail operations, and delivery management components to integrate seamlessly.
 - Provide a scalable and maintainable platform that can accommodate future growth in users and system functionality.

Information from Stakeholders

User Requirements

System scope, boundaries, and assumptions

- Assumptions
- 1. Assuming Woolworths has not implemented an order and delivery system and still relies on in-person grocery shopping.

Requirements

- ☐ Clearly defined functional requirements
- ☐ Clearly defined non-functional requirements
- ☐ Traceability between stakeholders and requirements

Requirement UML Modelling

- ☐ Use Case Diagram covering full system scope
- ☐ Minimum of five fully dressed Use Case Descriptions
- ☐ Correct use of include, extend, and generalisation relationships

Initial Structural Design and Implementation

- ☐ Initial UML Class Diagram
- ☐ Mapping between UML classes and planned C# classes
- ☐ Identification of anticipated design weaknesses prior to OOP application
- ☐ OOP-based implementation in C# that strictly conforms to the UML models.
- ☐ In the Class diagram, provide reasons for each component by writing a short summary of each.

Conclusion

References

1. Woolworths. (2024). Online Supermarket: Online Grocery Shopping & Free Recipes at countdown.co.nz. Countdown NZ Ltd.
<https://www.woolworths.co.nz/info/about-us/our-company>
2. O'Connell, J., & O'Connell, J. (2024, November 19). 1987 The Fresh Food People. Australian Food Timeline. <https://australianfoodtimeline.com.au/fresh-food-people/>

3. WOOLWORTHS GROUP LIMITED. (2025). Annual Report 2025.

<https://www.woolworthsgroup.com.au/content/dam/wwg/sustainability/reports/f25/Woolworths%20Group%20Annual%20Report%202025%20.pdf>

Appendices

- GitHub

(<https://github.com/ShawnLeeyz/Woolworths-Systems-Analysis-Design-and-Modelling-Project-I.git>)